

Bootstrap Methods

Bootstrap methods use computer simulation to reveal aspects of the sampling distribution for an estimator $\hat{\theta}$ of interest. With the power of modern computers the approach has broad applicability and is now a practical and useful tool for applied statisticians.

19.1 Introduction

To describe the bootstrap approach to inference, let $X_1, \dots, X_n$ be i.i.d. from some unknown distribution Q , and let

$$\mathbf{X} = (X_1, \dots, X_n)$$

denote all n observations. For now we proceed nonparametrically with Q an arbitrary distribution. Natural modifications when Q comes from a parametric family are introduced in Section 19.3. With Q arbitrary, a natural estimator for it would be the *empirical distribution*

$$\hat{Q} = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}.$$

Here δ_x represents a “point mass” at x , that assigns full probability to the point x , $\delta_x(\{x\}) = 1$, and zero probability to all other points, $\delta_x(\{x\}^c) = 0$. Then the estimator $\hat{Q}$ is a discrete distribution that assigns mass $1/n$ to each data point X_i , $1 \leq i \leq n$, and $\hat{Q}(A)$ is just the proportion of these values that lie in A :

$$\hat{Q}(A) = \frac{1}{n} \# \{i \leq n : X_i \in A\}.$$

Note that by the law of large numbers, $\hat{Q}(A) \xrightarrow{P} Q(A)$ as $n \rightarrow \infty$, supporting the notion that $\hat{Q}$ is a reasonable estimator for Q .

Suppose next that $\hat{\theta} = \hat{\theta}(\mathbf{X})$ is an estimator for some parameter $\theta = \theta(Q)$. Anyone using $\hat{\theta}$ should have interest in the distribution for the error $\hat{\theta} - \theta$, since

this distribution provides information about the bias, variance, and accuracy of $\hat{\theta}$. Unfortunately, this error distribution typically varies with Q , and because Q is unknown we cannot hope to know it exactly. Bootstrap methods are based on the hope or intuition that the true error distribution may be similar to the error distribution if the observations were sampled from $\hat{Q}$ instead of Q .

In principle, the error distribution with observations drawn from $\hat{Q}$ is a specific function of $\hat{Q}$, but exact calculations are generally intractable. This is where computer simulation plays an important role in practice. Given the original data $\mathbf{X}$, a computer routine draws a *bootstrap sample* $\mathbf{X}^* = (X_1^*, \dots, X_n^*)$, with the variables in this sample conditionally i.i.d. from $\hat{Q}$, so $\mathbf{X}^* | \mathbf{X} \sim \hat{Q}^n$. Note that since $\hat{Q}$ assigns mass $1/n$ to each observation, $X_1^*, \dots, X_n^*$ can be viewed as a random sample drawn with replacement from the set $\{X_1, \dots, X_n\}$. So these variables are very easy to simulate. If $\hat{\theta}^*$ is the estimate from the bootstrap sample,

$$\hat{\theta}^* = \hat{\theta}(\mathbf{X}^*) = \hat{\theta}(X_1^*, \dots, X_n^*),$$

then the distribution of $\hat{\theta}^* - \hat{\theta}$ is used to estimate the unknown distribution of the error $\hat{\theta} - \theta$. To be more precise, the estimate for the error distribution is the conditional distribution for $\hat{\theta}^* - \hat{\theta}$ given the original data $\mathbf{X}$. The following examples show ways this estimate for the error distribution might be used.

Example 19.1. Bias Reduction. Let

$$b = b(Q) = E_Q[\hat{\theta} - \theta]$$

denote the bias of an estimator $\hat{\theta} = \hat{\theta}(\mathbf{X})$ for a parameter $\theta = \theta(Q)$. If this bias were known, subtracting it from $\hat{\theta}$ would give an unbiased estimator. The true bias depends only on the error distribution. Substituting the bootstrap estimate for the true error distribution gives

$$\hat{b} = E[\hat{\theta}^* - \hat{\theta} | \mathbf{X}]$$

as the bootstrap estimate for the bias. Subtracting this estimate from $\hat{\theta}$ gives a new estimator $\hat{\theta} - \hat{b}$, generally less biased than the original estimator $\hat{\theta}$. Results detailing improvement are derived in the next section for a special case in which $\hat{\theta} = \bar{X}^3$ and θ is the cube of the mean of Q .

In practice, $\hat{b}$ would typically be computed by numerical simulation, having a computer routine draw multiple random samples, $\mathbf{X}_1^*, \dots, \mathbf{X}_B^*$, each from $\hat{Q}^n$. Letting $\hat{\theta}_i^* = \hat{\theta}(\mathbf{X}_i^*)$ denote the estimate from i th bootstrap sample $\mathbf{X}_i^*$, if the number of replications B is large, then by the law of large numbers $\hat{b}$ should be well approximated by the average

$$\frac{1}{B} \sum_{i=1}^B [\hat{\theta}_i^* - \hat{\theta}].$$

A natural assumption, relating the unknown parameter $\theta = \theta(Q)$ and the estimator $\hat{\theta}(\mathbf{X})$, is that the estimator has no error whenever proportions in the sample agree with probabilities from Q , which happens if $Q = \hat{Q}$. With this assumption,

$$\hat{\theta}(\mathbf{X}) = \theta(\hat{Q}), \quad (19.1)$$

and so $\hat{b} = b(\hat{Q})$. Hence, from a technical viewpoint, the bootstrap estimator here is found by plugging the empirical distribution $\hat{Q}$ into the functional of interest, $b(\cdot)$. This mathematical structure occurs generally and underlies various results in the literature showing that bootstrap methods perform well when functionals of interest are smooth in an appropriate sense.

Example 19.2. Confidence Intervals. Quantiles for $|\hat{\theta} - \theta|$ are useful in assessing the accuracy of $\hat{\theta}$, for if $q = q(Q)$ is the upper α th quantile¹ for the distribution of $|\hat{\theta} - \theta|$, then

$$P(\theta \in [\hat{\theta} - q, \hat{\theta} + q]) = 1 - \alpha.$$

The bootstrap estimator $\hat{q}$ for q would be the upper α th quantile for the conditional distribution of $|\hat{\theta}^* - \hat{\theta}|$ given $\mathbf{X}$. If this estimator is reasonably accurate, we expect that

$$P(\theta \in [\hat{\theta} - \hat{q}, \hat{\theta} + \hat{q}]) \approx 1 - \alpha,$$

so that $[\hat{\theta} - \hat{q}, \hat{\theta} + \hat{q}]$ is an approximate $1 - \alpha$ interval for θ . As in the bias example, $\hat{q}$ can be approximated numerically by simulation, still with random samples $\mathbf{X}_1^*, \dots, \mathbf{X}_B^*$ from $\hat{Q}^n$, again taking $\hat{\theta}_i^* = \hat{\theta}(\mathbf{X}_i^*)$. Then $\hat{q}$ could be approximated as the upper α th quantile for the list of values $|\hat{\theta}_i^* - \hat{\theta}|$, $i = 1, \dots, B$, generated in the simulation, or more formally as

$$\hat{q} \approx \inf \left\{ x : \frac{1}{B} \# \{i \leq B : |\hat{\theta}_i^* - \hat{\theta}| \leq x\} \geq 1 - \alpha \right\}.$$

Mathematically, the structure is much the same as the bias example, with the bootstrap estimator $\hat{q}$ obtained by plugging $\hat{Q}$ into $q(\cdot)$; that is, $\hat{q} = q(\hat{Q})$.

In practice, bootstrap confidence intervals can often be improved by modifying the approach and approximating the distribution of *studentized* errors, obtained by dividing the error $\hat{\theta} - \theta$ by an estimate of the standard deviation of $\hat{\theta}$. If $\tilde{q} = \tilde{q}(Q)$ is the upper α th quantile for the distribution of the absolute studentized error $|\hat{\theta} - \theta|/\hat{\tau}$, where $\hat{\tau} = \hat{\tau}(\mathbf{X})$ is an estimate for the standard deviation of $\hat{\theta}$, then

$$P(\theta \in [\hat{\theta} - \tilde{q}\hat{\tau}, \hat{\theta} + \tilde{q}\hat{\tau}]) = 1 - \alpha.$$

¹ For convenience, we assume that $\hat{\theta} - \theta$ has a continuous distribution. When this is not the case, some of the equations above may not hold exactly, but discrepancies will be quite small if masses for atoms of the error distribution are small.

The bootstrap estimator $\hat{\tilde{q}}$ for $\tilde{q}$ is the upper α th quantile for the conditional distribution of $|\hat{\theta}^* - \hat{\theta}|/\hat{\tau}^*$ given $\mathbf{X}$, where $\hat{\tau}^* = \hat{\tau}(\mathbf{X}^*)$. If this estimator is reasonably accurate, then

$$[\hat{\theta} - \hat{\tilde{q}}\hat{\tau}, \hat{\theta} + \hat{\tilde{q}}\hat{\tau}]$$

should be an approximate $1 - \alpha$ interval for θ . Again, the bootstrap estimate $\hat{\tilde{q}}$ can be computed numerically by simulation as the upper α th quantile for the list $|\hat{\theta}_i^* - \hat{\theta}|/\hat{\tau}_i^*$, $i = 1, \dots, B$, with $\hat{\theta}_i^*$ as before, and $\hat{\tau}_i^* = \hat{\tau}(\mathbf{X}_i^*)$.

To appreciate the value of studentizing, note that in various settings, including those detailed in the large-sample theory developed in Chapter 8, $(\hat{\theta} - \theta)/\hat{\tau}$ is approximately standard normal. If this is the case, the quantile $\tilde{q}$ is nearly independent of Q , making it easy to estimate. For instance, if the studentized error distribution happened to be exactly standard normal² for any Q , $\tilde{q}$ would always equal $z_{\alpha/2}$ and could be “estimated” perfectly, even without data.

19.2 Bias Reduction

In this section we explore a simple case of bias reduction where the performance of bootstrap estimators can be determined explicitly. Specifically, in Example 19.1 let

$$\theta = \theta(Q) = \mu^3,$$

where $\mu = EX_i$ is the mean of Q ,

$$\mu = \mu(Q) = \int x dQ(x).$$

The mean of $\hat{Q}$ is the average,

$$\mu(\hat{Q}) = \int x d\hat{Q}(x) = \overline{X} = \frac{1}{n} \sum_{i=1}^n X_i,$$

and so

$$\hat{\theta} = \theta(\hat{Q}) = \overline{X}^3.$$

To find the bias b we need $E\hat{\theta} = E\overline{X}^3$. Let $\sigma^2 = \text{Var}(X_i)$ and $\gamma = E(X_i - \mu)^3$. Since γ is the third cumulant for X_i , the third cumulant for $n\overline{X} = X_1 + \dots + X_n$ is $n\gamma$, and so

$$E(n\overline{X} - n\mu)^3 = n^3 E(\overline{X} - \mu)^3 = n\gamma.$$

Thus $E(\overline{X} - \mu)^3 = \gamma/n^2$. Also, $E(\overline{X} - \mu)^2 = \text{Var}(\overline{X}) = \sigma^2/n$. Using these identities,

² This could only happen in parametric situations.

$$\begin{aligned}
E\bar{X}^3 &= E(\bar{X} - \mu + \mu)^3 \\
&= E[\mu^3 + 3\mu^2(\bar{X} - \mu) + 3\mu(\bar{X} - \mu)^2 + (\bar{X} - \mu)^3] \\
&= \mu^3 + \frac{3\mu\sigma^2}{n} + \frac{\gamma}{n^2}.
\end{aligned} \tag{19.2}$$

So

$$b = b(Q) = E\hat{\theta} - \theta = E\bar{X}^3 - \mu^3 = \frac{3\mu\sigma^2}{n} + \frac{\gamma}{n^2}. \tag{19.3}$$

Because $\hat{b} = b(\hat{Q})$, we find the bootstrap estimate for b with the same calculations but with data drawn from $\hat{Q}$. The mean, variance, and third central moment of $\hat{Q}$ are

$$\bar{X} = \int x d\hat{Q}(x), \quad \hat{\sigma}^2 = \int (x - \bar{X})^2 d\hat{Q}(x) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2,$$

and

$$\hat{\gamma} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3.$$

Using these in (19.3),

$$\hat{b} = \frac{3\bar{X}\hat{\sigma}^2}{n} + \frac{\hat{\gamma}}{n^2}.$$

Subtracting this from $\bar{X}^3$, the “bias-reduced” estimator is

$$\hat{\theta} - \hat{b} = \bar{X}^3 - \frac{3\bar{X}\hat{\sigma}^2}{n} - \frac{\hat{\gamma}}{n^2}.$$

To see whether bootstrapping actually reduces the bias, we need to evaluate the mean of this new estimator. From (19.2) with $n = 1$, $EX_1^3 = \mu^3 + 3\mu\sigma^2 + \gamma$. Next, by symmetry, for any j ,

$$\begin{aligned}
EX_j \sum_{i=1}^n X_i^2 &= EX_1 \sum_{i=1}^n X_i^2 = EX_1^3 + \sum_{i=2}^n EX_1 X_i^2 \\
&= \mu^3 + 3\mu\sigma^2 + \gamma + (n-1)\mu(\mu^2 + \sigma^2).
\end{aligned}$$

Averaging over j ,

$$E\bar{X} \sum_{i=1}^n X_i^2 = \mu^3 + 3\mu\sigma^2 + \gamma + (n-1)\mu(\mu^2 + \sigma^2).$$

Finally, since

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2,$$

we have

$$\begin{aligned} E\bar{X}\hat{\sigma}^2 &= E\left[\frac{1}{n}\bar{X}\sum_{i=1}^n X_i^2 - \bar{X}^3\right] \\ &= \mu\sigma^2 + \frac{1}{n}(\gamma - \mu\sigma^2) - \frac{1}{n^2}\gamma. \end{aligned}$$

To find $E\hat{\gamma}$, note that

$$E(X_1 - \mu)^2(\bar{X} - \mu) = \frac{1}{n}E(X_1 - \mu)^3 = \frac{\gamma}{n},$$

and by symmetry

$$E(X_1 - \mu)(\bar{X} - \mu)^2 = \frac{1}{n}\sum_{i=1}^n E(X_i - \mu)(\bar{X} - \mu)^2 = E(\bar{X} - \mu)^3 = \frac{\gamma}{n^2}.$$

Using these and symmetry,

$$\begin{aligned} E\hat{\gamma} &= E(X_1 - \bar{X})^3 \\ &= E[(X_1 - \mu)^3 - 3(X_1 - \mu)^2(\bar{X} - \mu) \\ &\quad + 3(X_1 - \mu)(\bar{X} - \mu)^2 - (\bar{X} - \mu)^3] \\ &= \gamma\left(1 - \frac{3}{n} + \frac{2}{n^2}\right). \end{aligned}$$

Using these formulas, the mean of $\hat{\theta} - \hat{b}$ is

$$E[\hat{\theta} - \hat{b}] = \mu^3 + \frac{3}{n^2}(\mu\sigma^2 - \gamma) + \frac{6\gamma}{n^3} - \frac{2\gamma}{n^4}.$$

For large n the bias of this estimator is of order $1/n^2$ compared with a bias of order $1/n$ for $\hat{\theta}$. So bootstrap correction here typically improves the bias.³

19.3 Parametric Bootstrap Confidence Intervals

In this section we consider parametric models in which our data are i.i.d. from a distribution Q in some parametric family $\{Q_\lambda : \lambda \in \Lambda\}$. Knowing the marginal distribution lies in this family, $\hat{Q} = Q_{\hat{\lambda}}$, with $\hat{\lambda}$ the maximum likelihood estimator of λ , is a more natural estimator for Q than the empirical distribution used in earlier sections. With this modification, the bootstrap approach is essentially the same as before. If $\hat{\theta}$ is the maximum likelihood

³ If bias is the sole concern, there are unbiased estimators for μ^3 . The most natural one might be the U -statistic $\sum_{i < j < k} X_i X_j X_k / \binom{n}{3}$. Other resampling approaches, such as the jackknife, could also be used to reduce bias. See Problem 19.2.

estimator⁴ for a parameter $\theta = \theta(\lambda)$, and if $\mathbf{X}^*$ is a bootstrap sample with entries conditionally i.i.d. from $\hat{Q} = Q_{\hat{\lambda}}$, so

$$\mathbf{X}^* | \mathbf{X} \sim \hat{Q}^n,$$

then the error distribution for $\hat{\theta}$ would be estimated as the conditional distribution for $\hat{\theta}^* - \hat{\theta}$ given $\mathbf{X}$, with $\hat{\theta}^* = \hat{\theta}(\mathbf{X}^*)$.

As in Example 19.2, interval estimation for θ can be approached with an attempt to estimate the upper α th quantile $q = q(\lambda)$ for $|\hat{\theta} - \theta|$. The bootstrap estimate $\hat{q}$ for q is the upper α th quantile for the conditional distribution of $|\hat{\theta}^* - \hat{\theta}|$ given $\mathbf{X}$, and the approximate confidence interval based on $\hat{q}$ is

$$(\hat{\theta} - \hat{q}, \hat{\theta} + \hat{q}).$$

As before, studentizing the error distribution may give a more accurate confidence interval. If $\hat{\tau} = \hat{\tau}(\mathbf{X})$ is an estimate for the standard deviation of $\hat{\theta}$, and $\tilde{q} = \tilde{q}(\lambda)$ denotes the upper α th quantile for the distribution of the absolute studentized error $|\hat{\theta} - \theta|/\hat{\tau}$, then the bootstrap estimate $\hat{\tilde{q}}$ for $\tilde{q}$ is the upper α th quantile for the conditional distribution of $|\hat{\theta}^* - \hat{\theta}|/\hat{\tau}^*$ given $\mathbf{X}$, with $\hat{\tau}^* = \hat{\tau}(\mathbf{X}^*)$, and the associated approximate confidence interval is

$$(\hat{\theta} - \hat{\tilde{q}}\hat{\tau}, \hat{\theta} + \hat{\tilde{q}}\hat{\tau}).$$

If the functions $q(\cdot)$ or $\tilde{q}(\cdot)$ are tractable, quantile estimates $\hat{q}$ and $\hat{\tilde{q}}$ can be found by evaluation at the maximum likelihood estimator: $\hat{q} = q(\hat{\lambda})$ and $\hat{\tilde{q}} = \tilde{q}(\hat{\lambda})$. When this is not feasible or practical, these estimators can be approximated by bootstrap simulation. Specifically $\hat{q}$ or $\hat{\tilde{q}}$ would be approximated numerically as upper α th quantiles for the lists $|\hat{\theta}_i^* - \hat{\theta}|$, $i = 1, \dots, B$, or $|\hat{\theta}_i^* - \hat{\theta}|/\hat{\sigma}_i^*$, $i = 1, \dots, B$, with $\hat{\theta}_i^* = \hat{\theta}(\mathbf{X}_i^*)$, $\hat{\sigma}_i^* = \hat{\sigma}(\mathbf{X}_i^*)$, and $\mathbf{X}_i^*$, $i = 1, \dots, B$, conditionally i.i.d. from $\hat{Q}^n = Q_{\hat{\lambda}}^n$, given the original data $\mathbf{X}$.

To illustrate these ideas, consider interval estimation of the mean of a normal distribution. Taking $\lambda = (\theta, \sigma)$, $\hat{\lambda} = (\hat{\theta}, \hat{\sigma})$, with $\hat{\theta} = \bar{X}$ and

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Given $\mathbf{X}$, the resampled data $X_1^*, \dots, X_n^*$ are i.i.d. from $N(\bar{X}, \hat{\sigma}^2)$. Since $\hat{\theta}^* = (X_1^* + \dots + X_n^*)/n$, we have

$$\hat{\theta}^* | \mathbf{X} \sim N(\bar{X}, \hat{\sigma}^2/n) \text{ and } \hat{q} = z_{\alpha/2} \hat{\sigma} / \sqrt{n},$$

and the bootstrap confidence interval is

$$\left(\bar{X} - z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}} \right).$$

⁴ The maximum likelihood structure here replaces assumption (19.1).

Exploiting the independence of $\overline{X}$ and $\hat{\sigma}$, the coverage probability for this bootstrap confidence interval can be expressed as

$$\begin{aligned} P\left(\overline{X} - z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}} < \theta < \overline{X} + z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}}\right) &= P\left(-z_{\alpha/2} \frac{\hat{\sigma}}{\sigma} < \frac{\overline{X} - \theta}{\sigma/\sqrt{n}} < z_{\alpha/2} \frac{\hat{\sigma}}{\sigma}\right) \\ &= EP\left(-z_{\alpha/2} \frac{\hat{\sigma}}{\sigma} < \frac{\overline{X} - \theta}{\sigma/\sqrt{n}} < z_{\alpha/2} \frac{\hat{\sigma}}{\sigma} \mid \hat{\sigma}\right) \\ &= 1 - 2E\Phi\left(-\sqrt{z_{\alpha/2}^2 \frac{\hat{\sigma}^2}{\sigma^2}}\right). \end{aligned}$$

By Taylor expansion about $\hat{\sigma}^2/\sigma^2 = 1$,

$$\Phi\left(-\sqrt{z_{\alpha/2}^2 \frac{\hat{\sigma}^2}{\sigma^2}}\right) = \Phi(-z_{\alpha/2}) - \frac{1}{2} z_{\alpha/2} \phi(z_{\alpha/2}) \left(\frac{\hat{\sigma}^2}{\sigma^2} - 1\right) + o(\hat{\sigma}^2 - \sigma^2).$$

Taking expectations of this approximation⁵ and using

$$E\hat{\sigma}^2 = \frac{n-1}{n} ES^2 = \frac{n-1}{n} \sigma^2,$$

the coverage probability of the bootstrap confidence interval is

$$1 - \alpha - \frac{z_{\alpha/2} \phi(z_{\alpha/2})}{n} + o(1/n).$$

Because $\hat{\theta} = \overline{X}$ has standard deviation $\tau = \sigma/\sqrt{n}$, for the studentized approach we take $\hat{\tau} = \hat{\sigma}/\sqrt{n}$ and define $\tilde{q}(\lambda)$ as the upper α th quantile for $|\overline{X} - \theta|/\hat{\tau}$. If $S^2 = n\hat{\sigma}^2/(n-1)$ is the sample variance, then $T \stackrel{\text{def}}{=} \sqrt{n}(\overline{X} - \theta)/S \sim t_{n-1}$. Since

$$\frac{|\hat{\theta} - \theta|}{\hat{\tau}} = \frac{\sqrt{n}|T|}{\sqrt{n-1}z},$$

we have

$$\tilde{q}(\lambda) = \frac{\sqrt{n}t_{\alpha/2, n-1}}{\sqrt{n-1}}.$$

This quantile is independent of λ , and so $\hat{\tilde{q}} = \tilde{q}$ and the studentized bootstrap confidence interval is

$$(\hat{\theta} - \hat{\tilde{q}}\hat{\tau}, \hat{\theta} + \hat{\tilde{q}}\hat{\tau}) = (\overline{X} - t_{\alpha/2, n-1}S/\sqrt{n}, \overline{X} + t_{\alpha/2, n-1}S/\sqrt{n}).$$

This interval is the same as the usual t -confidence interval, so in this case studentizing works perfectly, giving a bootstrap confidence interval with coverage exactly $1 - \alpha$.

⁵ This can be justified using dominated convergence.

19.4 Nonparametric Accuracy for Averages

In this section we consider in some detail the performance of the nonparametric bootstrap when $\hat{\theta}$ is the sample mean $\bar{X}_n$, estimating $\theta = EX_i$. In this case error distributions can also be approximated using central limit theory, and it is of interest to see if a bootstrap approach does as well or better. Let $\sigma^2 = \text{Var}(X_i)$,

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2,$$

the variance of the empirical distribution $\hat{Q}_n$, and $\gamma = E(X_i - \theta)^3$. If

$$Z_n \stackrel{\text{def}}{=} \frac{\sqrt{n}(\hat{\theta} - \theta)}{\sigma},$$

then by the central limit theorem, $Z_n \Rightarrow N(0, 1)$ and $P(Z_n \leq x) \rightarrow \Phi(x)$. The next result shows that the error of this large-sample approximation is typically of order $1/\sqrt{n}$. Note that if Q is a lattice distribution, assigning probability one to a set of the form $\{a + bj : j \in \mathbb{Z}\}$, then this would have to be the case as the distribution for $\bar{X}_n$ would have to have jumps of order $1/\sqrt{n}$. The theorem assumes that Q is nonlattice.

Theorem 19.3. *If Q is nonlattice and $E|X_i|^3 < \infty$, then*

$$P(Z_n \leq x) = \Phi(x) - \frac{\gamma(x^2 - 1)}{6\sigma^3\sqrt{n}}\phi(x) + o(1/\sqrt{n})$$

as $n \rightarrow \infty$, uniformly in x .

A proof of this result is given in Appendix A.7.3. Let $Y_i = (X_i - \theta)/\sigma$, so that $Z_n = \sqrt{n}\bar{Y}_n$, let f be the characteristic function⁶ of Y_i , defined as

$$f(t) = Ee^{itY_i}, \quad t \in \mathbb{R},$$

and let f_n be the characteristic function of Z_n ,

$$f_n(t) = Ee^{itZ_n} = f^n(t/\sqrt{n}).$$

Using a smoothing lemma based on an inversion formula for characteristic functions (Lemma A.14), this theorem follows from two facts: first, that

$$f_n(t) = e^{-t^2/2} \left(1 - \frac{i\gamma t^3}{6\sigma^3\sqrt{n}} \right) + o(1/\sqrt{n}), \quad (19.4)$$

which follows by Taylor expansion of f using the first three moments of Y : $EY_i = 0$, $EY_i^2 = 1$, and $EY_i^3 = \gamma/\sigma^3$; and second, that for any $0 < \delta < c < \infty$,

⁶ See Appendix A.7.1.

$$\sup_{|t| \in [\delta, c]} |\mathfrak{f}(t)| < 1, \quad (19.5)$$

which follows from the nonlattice assumption.

A bootstrap approach to inference would approximate the distribution for Z_n by the conditional distribution for

$$Z_n^* \stackrel{\text{def}}{=} \frac{\sqrt{n}(\hat{\theta}^* - \hat{\theta})}{\hat{\sigma}_n}.$$

The following result, due to Singh (1981), shows that this conditional distribution is the same to $o_p(1/\sqrt{n})$, which represents an improvement over normal approximation.

Theorem 19.4. *If Q is nonlattice and $E|X_i|^3 < \infty$, then*

$$P(Z_n^* \leq x | \mathbf{X}) = \Phi(x) - \frac{\gamma(x^2 - 1)}{6\sigma^3\sqrt{n}}\phi(x) + o_p(1/\sqrt{n})$$

as $n \rightarrow \infty$, uniformly in x .

Proof (Sketch). If we define $Y_i^* = (X_i^* - \bar{X}_n)/\hat{\sigma}_n$, then $Z_n^* = \sqrt{n}\bar{Y}_n^*$, and given $\mathbf{X}$, $Y_1^*, \dots, Y_n^*$ are conditionally i.i.d. So the conditional structure here is identical to that in Theorem 19.3. Also,

$$E(Y_i^* | \mathbf{X}) = 0, \quad E(Y_i^{*2} | \mathbf{X}) = 1,$$

and

$$E(Y_i^{*3} | \mathbf{X}) = \frac{\hat{\gamma}_n}{\hat{\sigma}_n^3},$$

with

$$\hat{\gamma}_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^3.$$

So the first two conditional moments for Y_i^* are exactly the same as the corresponding moments of Y_i , and by the law of large numbers,

$$E(Y_i^{*3} | \mathbf{X}) = \hat{\gamma}_n / \hat{\sigma}_n^3 \xrightarrow{p} EY_i^3.$$

The same Taylor expansion argument used to show (19.4) gives

$$\mathfrak{f}_n^*(t) = e^{-t^2/2} \left(1 - \frac{i\gamma t^3}{6\sigma^3\sqrt{n}} \right) + o_p(1/\sqrt{n}),$$

with $\mathfrak{f}_n^*$ the conditional characteristic function for Z_n^* , $\mathfrak{f}_n^*(t) = E[e^{itZ_n^*} | \mathbf{X}]$.

The proof is completed, using the same argument used to prove Theorem 19.3, by showing that for any $0 < \delta < c < \infty$, there exists $\epsilon > 0$ such that

$$P\left(\sup_{|t| \in [\delta, c]} |\mathbf{f}^*(t)| > 1 - \epsilon\right) \rightarrow 0, \quad (19.6)$$

where $\mathbf{f}^*$ is the conditional characteristic function for Y_i^* , given by

$$\mathbf{f}^*(t) = E(e^{itY_i^*} \mid \mathbf{X}) = \frac{1}{n} \sum_{j=1}^n \exp\left[\frac{it(X_j - \bar{X}_n)}{\hat{\sigma}_n}\right], \quad t \in \mathbb{R}.$$

Using Theorem 9.2, our law of large numbers for random functions, it is not hard to show that

$$\sup_{|t| \in [\delta, c]} |\mathbf{f}(t) - \mathbf{f}^*(t)| \xrightarrow{P} 0,$$

and (19.6) then follows from (19.5). $\square$

Because the approximations in Theorems 19.3 and 19.4 hold uniformly in x ,

$$P[\sqrt{n}(\hat{\theta} - \theta) \leq x] = P(Z_n \leq x/\sigma) = \Phi(x/\sigma) - \frac{\gamma(x^2 - \sigma^2)}{6\sigma^5\sqrt{n}}\phi(x/\sigma) + o(1/\sqrt{n})$$

and

$$P[\sqrt{n}(\hat{\theta}^* - \hat{\theta}) \leq x \mid \mathbf{X}] = \Phi(x/\hat{\sigma}_n) - \frac{\gamma(x^2 - \sigma^2)}{6\sigma^5\sqrt{n}}\phi(x/\sigma) + o_p(1/\sqrt{n}).$$

Since $\hat{\sigma}_n - \sigma = O_p(1/\sqrt{n})$, by the delta method the leading terms in these approximations differ by $O_p(1/\sqrt{n})$. So in this case, bootstrap methods do a better job of approximating the distribution of the standardized variable Z_n than the distribution of the scaled error $\sqrt{n}(\hat{\theta} - \theta)$. Although the issues are somewhat different, this provides some support for the notion that studentizing generally improves the bootstrap performance.

The results above on the accuracy of bootstrap approximations can be extended in various ways. Perhaps the first thing worth noting is that Edgeworth expansions for distributions can be used to derive corresponding approximations, called *Cornish–Fisher expansions*, for quantiles. These expansions naturally play a central role in studying the performance of bootstrap confidence intervals.

Although the algebra is more involved, Edgeworth expansions can be derived to approximate distributions of averages of random vectors. And in principle these expansions lead directly to expansions for the distributions of smooth functions of averages. In his monograph, Hall (1992) uses this approach to study the performance of the bootstrap when $\hat{\theta}$ is a smooth function of averages. With suitable regularity, the discrepancy between the true coverage probability and the desired nominal value for the symmetric two-sided bootstrap confidence intervals described in Example 19.2 is of order $O(1/n)$ without studentizing, and of order $O(1/n^2)$ with studentizing.

As mentioned earlier, if $\theta = \theta(Q)$ and $\hat{\theta} = \theta(\hat{Q})$, then bootstrapping should work well if $\theta(\cdot)$ is suitably smooth. One regularity condition, studied in Bickel and Freedman (1981), is that $\theta(\cdot)$ is Gâteaux differentiable with the derivative representable as an integral. Such $\theta(\cdot)$ are often called *von Mises functionals*. Bickel and Freedman (1981) also give examples showing that bootstrapping can fail when $\theta(\cdot)$ is not smooth.

19.5 Problems

- Bootstrap methods can also be used to reduce bias in parametric estimation. As an example, suppose $X_1, \dots, X_n$ are i.i.d. from $N(\mu, 1)$, and consider estimating $\theta = \sin \mu$.
 - The maximum likelihood estimator, $\hat{\theta} = \sin \bar{X}$, is (for most μ) a biased estimator of θ . Derive an approximation for the bias of $\hat{\theta}$, accurate to $o(1/n^2)$ as $n \rightarrow \infty$.
 - Consider a parametric bootstrap approach to estimating the bias $b(\mu) = E_\mu \hat{\theta} - \theta$, in which, given $\mathbf{X} = (X_1, \dots, X_n)$, $X_1^*, \dots, X_n^*$ are conditionally i.i.d. from $N(\bar{X}, 1)$. Letting $\hat{b} = E[\hat{\theta}^* - \hat{\theta} | \mathbf{X}]$, derive an approximation for the bias of $\hat{\theta} - \hat{b}$, accurate to $o(1/n^2)$ as $n \rightarrow \infty$.
- Another resampling approach to inference is called the jackknife. Let $\hat{\theta}$ be an estimator for θ based on i.i.d. observations $X_1, \dots, X_n$, let $\hat{\theta}_{-i}$ be the estimator obtained omitting observation X_i from the data set, and define

$$\tilde{\theta}_i = n\hat{\theta} - (n-1)\hat{\theta}_{-i}, \quad i = 1, \dots, n,$$

called *pseudo-values* by Tukey.

- Let $\tilde{\theta}$ denote the average of the pseudo-values, and assume

$$E\hat{\theta} = \theta + \frac{a_1}{n} + \frac{a_2}{n^2} + o(1/n^2),$$

as $n \rightarrow \infty$. Derive an approximation for the bias of $\tilde{\theta}$ as $n \rightarrow \infty$, accurate to $o(1/n)$.

- Assume now that the observations X_i are random variables with a finite mean μ and variance $\sigma^2 \in (0, \infty)$, and that $\theta = A(\mu)$ and $\hat{\theta} = A(\bar{X})$ for some function A , with A' and A'' bounded and continuous. Show that

$$\frac{\sqrt{n}(\tilde{\theta} - \theta)}{\tilde{S}} \Rightarrow N(0, 1),$$

as $n \rightarrow \infty$, where $\tilde{S}$ is the sample standard deviation for the pseudo-values:

$$\tilde{S}^2 = \frac{1}{n-1} \sum_{i=1}^n (\tilde{\theta}_i - \tilde{\theta})^2.$$

3. Consider a parametric bootstrap approach to interval estimation with observations from a location family. Let $X_1, \dots, X_n$ be i.i.d. from an absolutely continuous distribution with density $f(x - \theta)$. Let $\hat{\theta}$ be the maximum likelihood estimator of θ , and define $q = q(\theta)$ as the upper α th quantile for the distribution of $|\hat{\theta} - \theta|$. The bootstrap estimate of q is $\hat{q} = q(\hat{\theta})$. Show that the associated confidence interval

$$(\hat{\theta} - \hat{q}, \hat{\theta} + \hat{q})$$

has exact coverage $1 - \alpha$.

4. Let q_α denote the upper α th quantile for $Z_n = \sqrt{n}(\bar{X}_n - \theta)/\sigma$ in Section 19.4. Use Theorem 19.3 to derive an approximation for q_α , accurate to $o(1/\sqrt{n})$ as $n \rightarrow \infty$.
5. Let $X_1, X_2, \dots$ be i.i.d. from a nonlattice distribution Q , with common mean μ , common variance σ^2 , and $E|X_i|^3 < \infty$; and let g be a twice continuously differentiable function with $g(\mu) \neq 0$. Use Theorem 19.3 to derive an approximation for

$$P[\sqrt{n}(g(\bar{X}_n) - g(\mu)) \leq x],$$

accurate to $o(1/\sqrt{n})$ as $n \rightarrow \infty$.

6. Let $X_1, X_2, \dots$ be i.i.d. absolutely continuous variables from a canonical exponential family with marginal density

$$f_\eta(x) = \exp\{\eta x - A(\eta)\}h(x), \quad x \in \mathbb{R},$$

for $\eta \in \Xi$. The maximum likelihood estimator of η based on the first n observations is then $\hat{\eta}_n = \psi(\bar{X}_n)$ with ψ the inverse of A' . Consider a parametric bootstrap approach to estimating the error distribution for $\hat{\eta}_n$. Given $\mathbf{X} = (X_1, \dots, X_n)$, let $X_1^*, \dots, X_n^*$ be conditionally i.i.d. with marginal density $f_{\hat{\eta}_n}$. Assume that the approximation derived in Problem 19.5 holds uniformly in some neighborhood of η , and use it to derive approximations for

$$P(\sqrt{n}(\hat{\eta}_n - \eta) \leq x)$$

and

$$P(\sqrt{n}(\hat{\eta}_n^* - \hat{\eta}) \leq x \mid \mathbf{X}),$$

both accurate to $o_p(1/\sqrt{n})$. Are these the same to $o_p(1/\sqrt{n})$?

